Math Inquirers

Extension Workbook

with answers

Level 6

Self-study | Ages 11-13



My name _____

About this book

Math Inquirers Extension Workbook, Level 6
Self-study practice with answers.

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Good to know

- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

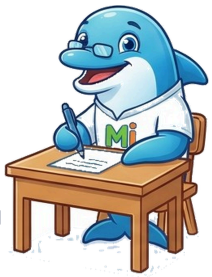
Contact

Math Inquirers, business WhatsApp +65 9880 5962. Email: pypteltd@gmail.com. Registered office: 66 Tannery Ln, Singapore 347805.

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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Code the pattern	Not in this proof edition				☐
2	Prime hunters	Not in this proof edition				☐
3	Below zero	6-7	8-9	10 and 12	11	☐
4	Algebra machines	Not in this proof edition				☐
5	Every possible way	Not in this proof edition				☐
6	Fair or unfair?	Not in this proof edition				☐
7	Turtle geometry	Not in this proof edition				☐
8	Secret codes	Not in this proof edition				☐
9	Data detectives	Not in this proof edition				☐
10	Showcase challenge	Not in this proof edition				☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

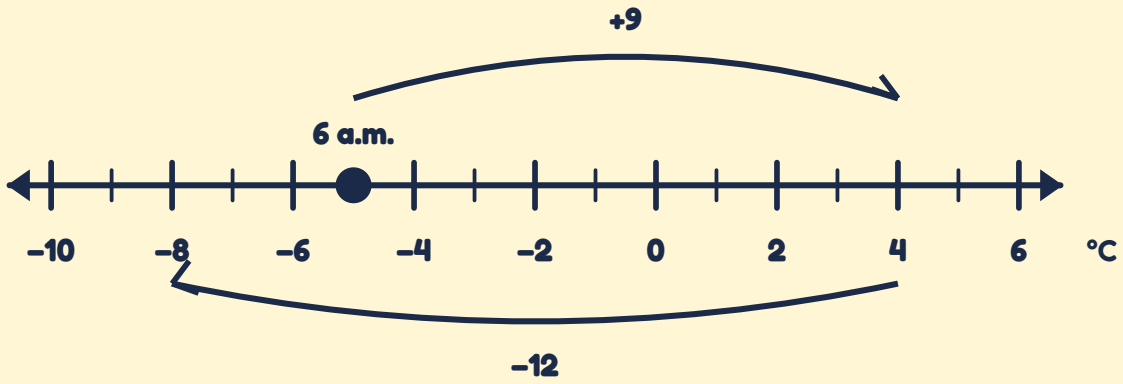
Learn it: Negative numbers

My goal. I can order negative numbers and add and subtract by jumping along a number line. I can follow a variable that goes below 0 in code.

Numbers below 0 have a minus sign. On a number line, greater numbers are to the right. To add, jump right. To subtract, jump left. When a jump crosses 0, split it at 0.

Worked example

It is –5 °C at 6 a.m. By noon it rises 9 degrees. By night it falls 12 degrees. What is the temperature at night?



1. **Start at –5. Rising moves right. 5 steps reach 0, then 4 more steps: 4 °C at noon.**
2. **Falling moves left. From 4, 4 steps reach 0, then 8 more steps.**
3. **The night temperature is –8 °C.**
4. **Check by adding: –5 + 9 – 12 = –8.**

Answer: The temperature at night is –8 °C.

How to add and subtract with negative numbers

1. Mark the start on the number line.
2. Adding moves right. Subtracting moves left.
3. If the jump crosses 0, split it at 0.
4. Check by jumping back to the start.

Your turn

Answer at the back of the book.

Try It is –2 °C. It falls 7 degrees. What is the temperature now? ☐ °C

Start here

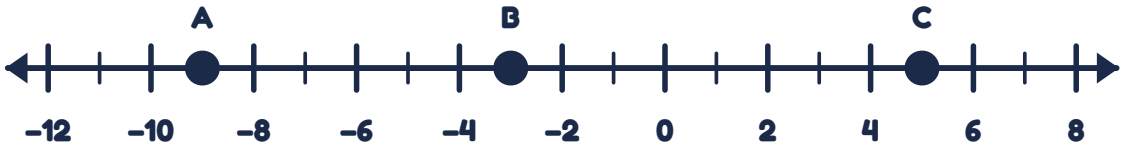
Do these first. Take your time.

1 **Put the numbers in order. Begin with the smallest.**

(a) 2, –6, –11, 0, –3, 8 gives _____

(b) Circle > or <: –4 > < –10 0 > < –2 –15 > < 1

2 **Read the number line. Each mark is 1 step.**



(a) A = ☐ B = ☐ C = ☐

(b) From A to B is ☐ steps. From B to C is ☐ steps.

3 **The thermometer shows 3 readings on one day.**

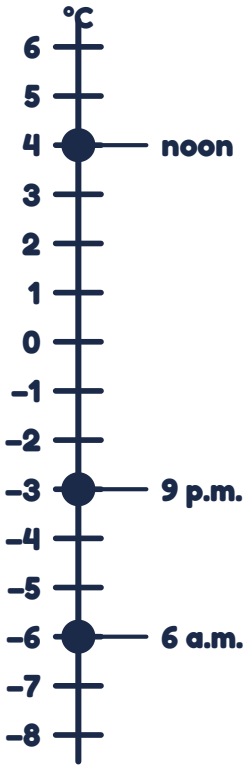
(a) At 6 a.m.: ☐ °C

(b) At noon: ☐ °C

(c) At 9 p.m.: ☐ °C

(d) From 6 a.m. to noon it rose ☐ degrees.

(e) From noon to 9 p.m. it fell ☐ degrees.



Start here

Keep going. Check your work.

4 Find the value of each one. Use a number line to help.

(a) $-6 + 10 =$ ☐ (b) $3 - 9 =$ ☐ (c) $-7 - 5 =$ ☐

(d) $-4 + 15 =$ ☐

5 Find the missing number.

(a) $-5 +$ ☐ $= 6$ (b) $4 -$ ☐ $= -3$ (c) ☐ $+ 8 = 2$

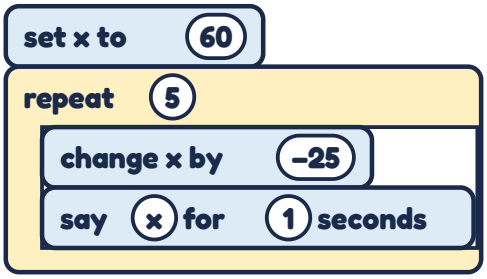
(d) ☐ $- 5 = -9$

6 How far is it from the first number to the second? Count the steps, splitting at 0.

(a) -9 to 4: ☐ steps (b) -6 to -1: ☐ steps

(c) -14 °C to 31 °C: ☐ degrees

7 Trace the code. The sprite changes x, then says it.



Loop	x said
1	
2	
3	
4	
5	

(a) The 3rd number it says is ☐.

(b) The last number it says is ☐.

(c) It says ☐ numbers that are below 0.

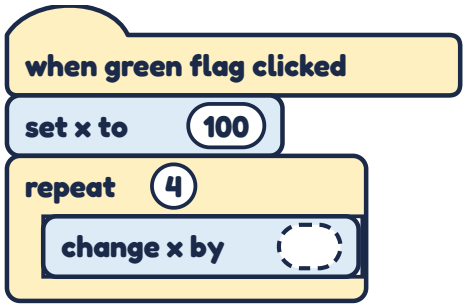
Start here

Nearly there. You can do it.

8 **A sprite starts at x = 100. Fill in the blocks.**

(a) It ends at x = 0. change x by ☐

(b) With repeat 8 and change x by (–20), it ends at x = ☐.



9 **A balance of –15 dollars means Ali owes 15 dollars. Ali has 25 dollars.**

(a) He buys a game for 40 dollars. His balance is ☐ dollars.

(b) Then he is given 30 dollars. His balance is ☐ dollars.

(c) To go from –15 dollars up to 20 dollars, he needs ☐ dollars.

10 **In a car park the floors are L3, L2, L1, B1, B2 and B3. There is no floor 0. Count the floors one by one.**

L3	L2	L1	B1	B2	B3

(a) A car goes down from L3 to B2: ☐ floors.

(b) A car goes up from B3 to L1: ☐ floors.

(c) A car goes down from L2 to B1: ☐ floors.

11 **Think about how far each number is from 0.**

(a) Circle the number closest to 0: –9 5 –3 7 –4

(b) Which is further from 0? –8 6

Think harder

Hints are upside down. Try first!

1 **Beyond Level 6: Subtracting a negative number.**
To find 6 – (–4), count the steps from –4 up to 6.

(a) 6 – (–4) = ☐ (b) –2 – (–5) = ☐

(c) So taking away –4 gives the same result as adding ☐.

Number line: _____

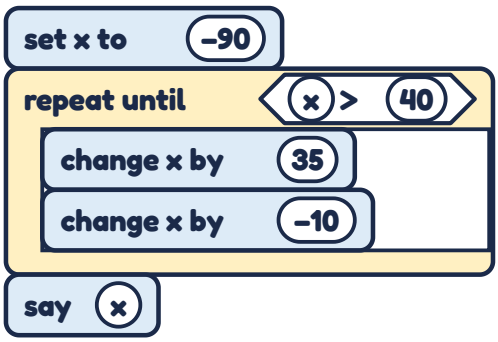
_____

Hint. How far is it from the second number up to the first number?

2 **A sprite starts at x = –90. The loop stops as soon as x is greater than 40 at the start of a repeat.**

(a) The loop repeats ☐ times.

(b) The sprite says ☐.



Hint. Write x after each repeat. Is it more than 40 yet?

3 **On Monday at noon it was 3 °C. On Tuesday it was 11 degrees colder than Monday. On Wednesday it was 6 degrees warmer than Tuesday. On Thursday it was –4 °C.**

(a) Tuesday: ☐ °C (b) Wednesday: ☐ °C

(c) From Monday to Thursday it fell ☐ degrees.

Number line: _____

_____

Hint. Draw a number line. Mark Monday first, then each change.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Two divers: A is at −12 m and goes down 4 m every minute. B is at −40 m and comes up 6 m every minute.**

(a) After 3 minutes, A is at ☐ m and B is at ☐ m.

(b) B is first above A after minute ☐.

(c) After 5 minutes they are ☐ m apart.

Working: _____

_____

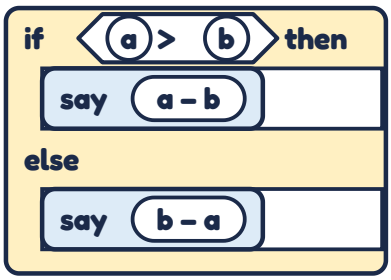
Hint. Make a table: minute, depth of A, depth of B.

2 **This code uses a and b. Run it for each pair of numbers.**

(a) a = −8 and b = 5. It says ☐.

(b) a = −3 and b = −9. It says ☐.

(c) a = −6 and b = −6. It says ☐.



Hint. Is a greater than b? Choose the right branch first.

3 **Beyond Level 6: A calendar with no year 0.**
There is no year 0. The year after 1 BC is AD 1. Count the years one by one.

(a) From 5 BC to 1 BC is ☐ years.

(b) From 3 BC to AD 4 is ☐ years.

(c) A person born in 10 BC dies in AD 12. They are ☐ years old.

Working: _____

_____

Hint. A number line with 0 would be 1 step too short.

✔ Check yourself

No hints. Try each one on your own.

1 **–9 + 14 =** ☐ **6 – 13 =** ☐

Working: _____

2 **Put in order, smallest first: 4, –7, –2, 0**

_____

Working: _____

3 **It is –6 °C. It rises 11 degrees, then falls 8 degrees. It is now** ☐ **°C.**

Working: _____

4 **A sprite starts at x = 20. It runs repeat 3 [change x by –15]. Then x =** ☐ **.**

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Code the pattern	☐	☐	☐	☐	/ 4	
2	Prime hunters	☐	☐	☐	☐	/ 4	
3	Below zero	☐	☐	☐	☐	/ 4	
4	Algebra machines	☐	☐	☐	☐	/ 4	
5	Every possible way	☐	☐	☐	☐	/ 4	
6	Fair or unfair?	☐	☐	☐	☐	/ 4	
7	Turtle geometry	☐	☐	☐	☐	/ 4	
8	Secret codes	☐	☐	☐	☐	/ 4	
9	Data detectives	☐	☐	☐	☐	/ 4	
10	Showcase challenge	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 6.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 3 · Below zero

Your turn
page 6

Answer: −9
Why: Falling moves left. From −2 jump left 7 steps: −2 + (−7) = −9. Check: from −9 jump right 7 to reach −2.

START HERE

1
page 7

Answer: (a) −11, −6, −3, 0, 2, 8 **(b)** >, >, <
Why: The smallest number is the furthest left on the number line: −11, −6, −3, 0, 2, 8. (b) On the number line, the number on the right is greater. −4 is to the right of −10, 0 is right of −2, and −15 is left of 1.

2
page 7

Answer: (a) −9, −3, 5 **(b)** 6, 8
Why: The labelled numbers are every 2 steps. A is 1 step right of −10, so −9. B is 1 step left of −2, so −3. C is 1 step left of 6, so 5. From −9 to −3 is 6 steps and from −3 to 5 is 8 steps (3 steps to 0, then 5 more).

3
page 7

Answer: (a) −6 **(b)** 4 **(c)** −3 **(d)** 10 **(e)** 7
Why: Read each mark: 6 a.m. is −6 °C, noon is 4 °C and 9 p.m. is −3 °C. From −6 up to 4 is 6 steps to 0 and 4 more: 10 degrees. From 4 down to −3 is 4 steps to 0 and 3 more: 7 degrees.

4
page 8

Answer: (a) 4 **(b)** −6 **(c)** −12 **(d)** 11
Why: (a) From −6 jump right 10: 4 steps to 0, then 6 more: 4. (b) From 3 jump left 9: 3 steps to 0, then 6 more: −6. (c) From −7 jump left 5: −12. (d) From −4 jump right 15: 4 steps to 0, then 11 more: 11.

5
page 8

Answer: (a) 11 **(b)** 7 **(c)** −6 **(d)** −4
Why: (a) From −5 to 6 is 5 steps to 0 and 6 more: 11. (b) 4 − 7 = −3, so 7. (c) −6 + 8 = 2, so the missing number is −6. (d) −4 − 5 = −9, so the missing number is −4.

6
page 8

Answer: (a) 13 **(b)** 5 **(c)** 45
Why: (a) From −9 to 4: 9 steps to 0 and 4 more is 13. (b) From −6 to −1: 5 steps, no need to cross 0. (c) From −14 to 31: 14 steps to 0 and 31 more is 45.

7
page 8

Answer: (a) −15 **(b)** −65 **(c)** 3
Why: x starts at 60. Loop 1: 60 − 25 = 35. Loop 2: 10. Loop 3: −15. Loop 4: −40. Loop 5: −65. The 3rd number said is −15. The last is −65. The numbers below 0 are −15, −40 and −65: 3 of them.

8
page 9

Answer: (a) −25 **(b)** −60
Why: (a) 4 equal moves must go from 100 down to 0. That is 100 left in 4 moves, so each move is −25. (b) 8 moves of −20 make −160, and 100 − 160 = −60.

9
page 9

Answer: (a) −15 **(b)** 15 **(c)** 35
Why: (a) 25 − 40 = −15. (b) −15 + 30 = 15. (c) From −15 up to 20: 15 dollars to reach 0, then 20 more, 35 dollars.

10 page 9

Answer: (a) 4 (b) 3 (c) 2
Why: (a) L3, L2, L1, B1, B2: that is 4 moves down. (b) B3, B2, B1, L1: 3 moves up. (c) L2, L1, B1: 2 moves down. A number line with 0 would give one move too many.

11 page 9

Answer: (a) −3 (b) −8
Why: (a) The distances from 0 are 9, 5, 3, 7 and 4. −3 is only 3 steps from 0. (b) −8 is 8 steps from 0 and 6 is 6 steps from 0, so −8 is further.

THINK HARDER

1 page 10

Answer: (a) 10 (b) 3 (c) 4
Why: (a) 6 − (−4): from −4 to 6 is 10 steps. (b) −2 − (−5): from −5 to −2 is 3 steps. (c) In both, subtracting a negative number gave the same answer as adding the matching positive number: 6 + 4 and −2 + 5.

2 page 10

Answer: (a) 6 (b) 60
Why: Each repeat changes x by 35 − 10 = 25. After the repeats x is −65, −40, −15, 10, 35, 60. After 5 repeats x = 35, which is not greater than 40, so it repeats a 6th time. It stops with x = 60.

3 page 10

Answer: (a) −8 (b) −2 (c) 7
Why: Tuesday: 3 − 11 = −8. Wednesday: −8 + 6 = −2. Thursday is −4, which is 2 below Wednesday. From Monday (3) to Thursday (−4): 3 steps to 0 and 4 more, so it fell 7 degrees.

GO FURTHER

1 page 11

Answer: (a) −24, −22 (b) 3 (c) 22
Why: After 3 minutes A is at −12 − 12 = −24 and B is at −40 + 18 = −22. B is now above A. After 2 minutes A is −20 and B is −28, so B is still below. After 5 minutes A is −32 and B is −10: the gap is 22 m.

2 page 11

Answer: (a) 13 (b) 6 (c) 0
Why: It says how far apart a and b are. a = −8 and b = 5: a is not greater, so it says 5 − (−8) = 13. a = −3 and b = −9: a is greater, so it says −3 − (−9) = 6. The program always says the gap between a and b, whichever is greater.

3 page 11

Answer: (a) 4 (b) 6 (c) 21
Why: (a) 5 BC, 4 BC, 3 BC, 2 BC, 1 BC: 4 steps. (b) 3 BC to 2 BC, 1 BC, AD 1, AD 2, AD 3, AD 4: 6 steps. (c) 10 BC to 1 BC is 9 steps, 1 step to AD 1, then 11 steps to AD 12: 21 years.

CHECK YOURSELF

1 page 12

Answer: 5, −7
Why: From −9 jump right 14: 9 steps to 0 and 5 more is 5. From 6 jump left 13: 6 steps to 0 and 7 more is −7.

2 page 12

Answer: −7, −2, 0, 4
Why: On a number line, from left to right: −7, −2, 0, 4.

3
page 12

Answer: −3
Why: −6 + 11 = 5. Then 5 − 8 = −3.

4
page 12

Answer: −25
Why: 3 moves of −15 make −45. 20 − 45 = −25.